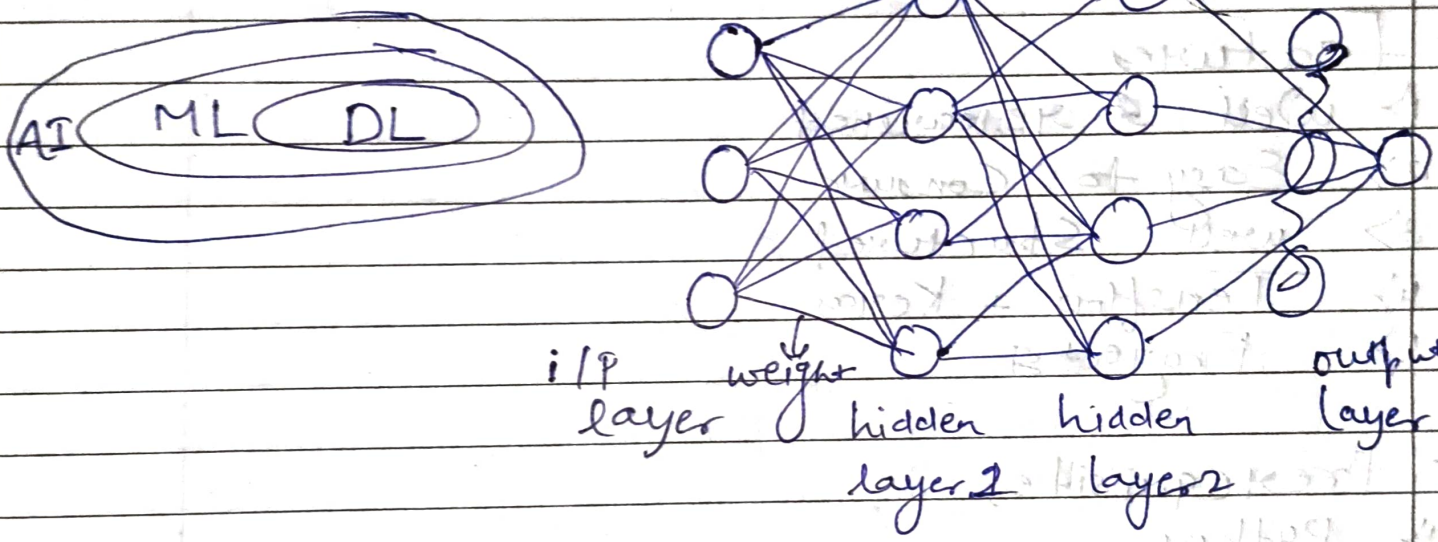
SdS

Day-2

→ What Is Deep learning?

- Deep learning is a subfield of AI and ML that is inspired by the structure of human brain.

- Deep learning algorithm to draw similar conclusions as humans would by continually analyzing data with a given logical structure called Neural Network.



ANN →
CNN → Best for image data
RNN → " " speech / textual data
GAN → text / image generation.

Q) why dl getting so famous?
1) Applicability → Covers very large domain
2) performance → State-of-the-art

SRS

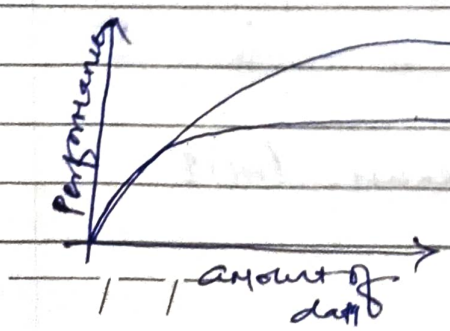
→ DL is part of a broader family of ML methods based on ANN with representation learning

DL algorithm uses multiple layers to progressively extract higher level features from raw input. for example, in image processing, lower layers may identify edges while higher layer may identify the concepts relevant to a human such as digits or letters or faces.

→ Deep learning VS Machine learning

	DL	ML
1) Data Dependency	High	Low
Less data ML wins but	Data ↑ DL performance ↑	
2) Hardware Dependency		
• Matrix Multiplication → GPUs required		
3) Training time	Very high	low
Prediction time	fast	slow
4) Feature Selection	Automatically	Manually
5) Interpretability (How)	Low	High

• DL can't Replace ML
ML - सूई
DL - तलवार



SDS

→ Why Now?
1960's Alan Turing started
Breakthrough 2012

1) Hardware
2) Architecture
3) Data sets → data hungary
4) Frameworks
5) Community

Start → CPU
Large → GPU, TPU
Smartphone → Mobile CPU, GPU, DSP, NPU
Smart watch → Edge TPU, NPU

FPGA →
↳ Xilinx

ASIC →
TPU → tensor processing unit (google)
Edge TPU → Drone, watch, glasses
NPU → Neural processing unit

→ Libraries
Tensorflow. (google) 2015
• very powerful → very difficult
TF Keras + TF

PyTorch (facebook) 2016
AI Researcher
2018 - Caffe 2
pytorch

pytorch → Tensorflow
↳ Drop down GUI

SdS

Image classification → ResNet
Text " → BERT
Image " → UNET
Image Generation → pix2 Pix
Object detection → YOLO
Speech generation → wave net

→ Transfer learning → In this learning we pull other Neural Network and use for our ~~purpose~~ purpose

→ Community

1960 → 2011